GameBuddy v4.1

Official User Manual & Optimization Guide — HackVerse Gaming 2.0

Welcome to **GameBuddy**, the ultimate Windows PC gaming optimization and repair suite. Designed to bridge the gap between complex system diagnostics and 1-click solutions, GameBuddy acts as your personal command center. Whether you are dealing with mysterious engine crashes, unoptimized runtime environments, or corrupted shader caches causing stuttering, this utility provides absolute control over your gaming experience.

Phase 1: Installation & Quick Start

How to Download and Run

GameBuddy is built as a portable, standalone executable. It does not require a complex installation wizard. Follow these steps to get started safely:

1. **Download:** Visit the official HackVerse GitHub Releases page and download the `GameBuddy_v4.1_Windows.zip` file.
2. **Extract:** Right-click the downloaded ZIP archive and select *Extract All...* to unpack the contents to a folder on your Desktop or Documents.
3. **Administrator Privileges:** Right-click `GameBuddy_v4.1.exe` and select **Run as Administrator**. *Note: Running as Admin is strictly required for the application to perform Winsock resets, DNS flushes, and Visual C++ installations.*
4. **First Boot:** Upon launch, GameBuddy will automatically create a secure workspace in your Windows `%APPDATA%/GameFixHub` folder to store backups and logs.

Understanding SmartScreen Warnings

Because GameBuddy interacts with deep Windows network catalogs and automated file diagnostics, Windows Defender SmartScreen or Google Chrome may occasionally flag the executable as "Unrecognized" or a false positive. GameBuddy is 100% open-source and free of telemetry. If blocked by Windows SmartScreen, click **More Info** and then **Run Anyway**.

Module 1: The Verified Game Library

GameBuddy eliminates the clutter of standard launchers by aggregating only locally verified installations.

Auto-Discovery & Cover Art

When you click **Rescan** on the top bar, GameBuddy runs a background worker thread that searches your drives (C:, D:, E:) for Steam, Epic Games, MS Store, and GOG directories. It reads `.acf` manifest files to prove the game exists locally before adding it to the dashboard.

- **Missing a game?** Click the + *Add Installed Game Folder* button to manually point the engine to custom installation directories or repacks.
- **Real Covers:** The app reaches out to official Content Delivery Networks (CDNs) to download high-resolution cover art directly into your local cache.



Module 2: Advanced Diagnostics Engine

Hunting Crash Logs

Standard game launchers fail to tell you *why* a game crashed. GameBuddy's Deep Diagnostics Engine actively hunts through the selected game's directory for proprietary engine crash logs. To use this feature:

1. Select a game from the **Library** tab.
2. Navigate to the **Diagnostics** tab on the left sidebar.
3. Click **Deep-Scan Engine Logs**.

The scanner reads `Player.log` (Unity), `CrashContext.xml` (Unreal Engine), and `launcher.log` (Rockstar). It will highlight specific hex codes or missing DLLs in the Results pane, separating false alarms from critical launch failures.

Module 3: System Health Check

Rig Score & Telemetry

The System Check tab is your PC's physical examination room. Rather than guessing if your PC is ready for a heavy session, this module queries Windows Management Instrumentation (WMI) to generate a live **Rig Score** (0-100%).

- **Overview:** Calculates overall health based on missing critical files, free disk space, and network reachability.
- **Hardware Sub-Tab:** Displays hard-coded CPU architecture, physical memory loads, and the exact active GPU engine model currently processing graphics.
- **Software Sub-Tab:** Verifies the presence of crucial gaming dependencies, specifically checking `System32` for Visual C++ 2015-2022 Redistributable files (like `vcruntime140.dll`).

Module 4: Network Doctor

Resolving Connection Drops

If you are experiencing "Cannot connect to server" or infinite loading screens in multiplayer titles (like Valorant or Apex Legends), the Network tab provides instant remediation. The tool explicitly tests HTTPS reachability without executing bandwidth-heavy download tests.

Diagnostic Test	Target System	Status Output
DNS Resolution	Local ISP Node	Working or Failed
Google TCP (8.8.8.8)	Global Routing	Reachable or Timeout
Flush DNS Cache	Windows IPConfig	Clears stale IP addresses instantly.
Reset Winsock	Network Adapters	Requires PC Restart. Fixes corrupted sockets.

Module 5: Guided Repairs & Backups

Safe Configuration Manifests

Modding games or tweaking graphics `.ini` files can easily break a stable installation. GameBuddy includes a true cryptographic backup system.

1. Navigate to the **Repairs & Backup** tab and click *Create Backup*.
2. Select the game. GameBuddy isolates and zips only configuration formats (`.ini, .cfg, .json, .sav, .xml`) to avoid duplicating massive texture files.
3. The system injects a `manifest.json` into the archive. This ensures that when you click the **Restore** sub-tab, the files are overwritten into the exact correct path.

All backups are securely stored in your user profile at `AppData\Roaming\GameFixHub\Backups`, keeping them safe even if you uninstall the main game.

Module 6: Pre-Game FPS Booster

Clearing Corrupted Shaders

Micro-stuttering in modern games (especially Unreal Engine 4/5 titles) is heavily tied to full or corrupted shader caches. Located in the *Guided Fixes* section, the **Clean Shader Cache** tool performs targeted maintenance:

- Navigates silently to your `AppData/Local/NVIDIA/DXCache` and `AMD/DxCache` folders.
- Deletes all unlocked, legacy compiled shaders that are taking up space or causing conflict.
- Forces the game to compile a fresh, optimized pipeline on your next launch, significantly smoothing out frame-time spikes.

Frequently Asked Questions (FAQ)

Why does the app stay on "Scanning drives..."?

If you have multiple terabytes of game installations across several hard drives, the initial verification scan can take time. GameBuddy verifies the existence of actual executables rather than just reading registry keys. You can safely click **Cancel Scan** at the top right of the dashboard at any time to halt the background thread and browse the games found so far.

Is GameBuddy Open Source?

Yes. The entire codebase is available for public audit on the official GitHub repository. We believe system utilities should be completely transparent. Developers are welcome to review the threading, file paths, and network calls directly in the Python source files.

Community & Support

Having trouble fixing a specific game, or want to request a new feature for the next version?

- **GitHub Issues:** Submit bug reports directly on the repository page.
- **YouTube:** Check out the official video tutorials on the **HackVerse Gaming 2.0** channel for visual walkthroughs of the advanced repair tools.